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INSUREWELL full-stack

# Renewal Reminder PDF Feature Change Set Report

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**Environment:** Java 17, Spring Boot 3.1, React 18, Playwright Node, ReportLab, python-pptx

**Status:** Features fully implemented, verified, and E2E tested.

This report provides a comprehensive summary of the implementation and validation of the Downloadable PDF Renewal Reminder feature. It captures the backend service changes (utilizing Apache PDFBox), React frontend integrations, and complete end-to-end automated testing results.

# 1. Executive Summary & Automated Test Results

The Renewal Reminder feature adds robust functionality to the InsureWell policy dashboard. When an active policy is close to renewal (or has expired), the system displays a prominent warning banner to the user. Additionally, users can download an official, professional PDF renewal reminder document directly from either the banner or the policy details card. The PDF generation is handled dynamically on the Spring Boot backend using Apache PDFBox, providing an instant, high-quality download experience.

## Automated Test Verification Summary

The feature was rigorously validated using both backend JUnit/MockMvc controller tests and frontend Playwright E2E integration tests. All tests completed successfully with a 100% pass rate.

| Test Suite                   | Framework / Tool         | Tests Run | Passed | Failures | Result Status |
|------------------------------|--------------------------|-----------|--------|----------|---------------|
| Backend PolicyControllerTest | JUnit 5 / Spring MockMvc | 2         | 2      | 0        | PASSED        |
| Frontend Navigation & E2E    | Playwright               | 23        | 23     | 0        | PASSED        |

**Backend Unit Verification:** PolicyControllerTest.java covers the dynamic generation of reminders, ensuring PDF endpoints return correct content types, valid PDF headers, and appropriate error states for invalid policy IDs.

**Frontend E2E Verification:** dashboard.spec.ts verifies navigation, policy modal forms, policy statistics updates, conditional banner rendering for POL-2024-001 and POL-2023-009, lack of banner for POL-2024-002, and successful triggers of the dynamic download workflow.

## 2. Implementation Details & File Changes

The change set represents a highly-coordinated integration between the Java Spring Boot backend and the React frontend. Below is a detailed file-by-file summary of the modifications implemented:

| File Path                      | Ecosystem | Primary Responsibility                                                                           |
|--------------------------------|-----------|--------------------------------------------------------------------------------------------------|
| pom.xml                        | Backend   | Adds org.apache.pdfbox dependency for dynamic PDF document creation.                             |
| PolicyController.java          | Backend   | Exposes the GET endpoint /api/policies/{id}/reminder-pdf to return the generated PDF file.       |
| RenewalReminderPdfService.java | Backend   | Implements the core PDFBox generation logic, embedding logos, metadata, policy details, and a    |
| PolicyControllerTest.java      | Backend   | Verifies the controller endpoint responses, headers, and exception handling.                     |
| Claims.js                      | Frontend  | Minor adjustments to support claims table filtering by active policies.                          |
| Dashboard.js                   | Frontend  | Integrates the RenewalReminderBanner, conditional download cards, and PDF API dispatchers.       |
| RenewalReminderBanner.js       | Frontend  | Builds the user-facing alert UI, displaying orange/red warning banners depending on expiration s |
| Dashboard.css                  | Frontend  | Implements response layouts, banner warning palettes, and customized download buttons.           |
| dashboard.spec.ts              | Frontend  | Includes full E2E Playwright coverage verifying banner states and user download flows.           |

### 3. Highlighted Implementation Code Snippets

Below are key extracts from the backend controller and service responsible for generating and returning the dynamic PDF reminders:

#### PolicyController.java - Endpoint Exposition

```
@GetMapping("/{id}/reminder-pdf")
public ResponseEntity getRenewalReminderPdf(@PathVariable Long id) {
    try {
        byte[] pdfBytes = reminderPdfService.generateRenewalReminderPdf(id);
        HttpHeaders headers = new HttpHeaders();
        headers.setContentType(MediaType.APPLICATION_PDF);
        headers.setContentDispositionFormData("attachment", "renewal-reminder-" + id + ".pdf");
        headers.setCacheControl("must-revalidate, post-check=0, pre-check=0");
        return new ResponseEntity<>(pdfBytes, headers, HttpStatus.OK);
    } catch (ResourceNotFoundException e) {
        return ResponseEntity.notFound().build();
    } catch (Exception e) {
        return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR).build();
    }
}
```

#### RenewalReminderPdfService.java - PDFBox Canvas Drawing

```
try (PDDocument document = new PDDocument()) {
    PDPage page = new PDPage(PDRectangle.LETTER);
    document.addPage(page);
    try (PDPageContentStream contentStream = new PDPageContentStream(document, page)) {
        // Draw Brand Header Accent Bar
        contentStream.setNonStrokingColor(30, 61, 89); // Deep Navy
        contentStream.addRect(0, 750, 612, 42);
        contentStream.fill();

        // Draw Document Title
        contentStream.beginText();
        contentStream.setFont(PDType1Font.HELVETICA_BOLD, 22);
        contentStream.newLineAtOffset(50, 680);
        contentStream.showText("POLICY RENEWAL REMINDER");
        contentStream.endText();
        ...
    }
}
```

## 4. User Interface Verification & Screenshots

This section highlights the visual layout of the implemented feature as captured by the Playwright end-to-end driver.

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DashboardClaims

### Policy Dashboard

6 policies on your account

Alex JohnsonPOL-2024-001

Maria GarciaPOL-2024-002

David ChenPOL-2023-009

Jane Doe

 **Policy Expiring Soon**  
This policy will expire in 168 days. Action required.

Renew

Download Reminder

**InsureWell Premium Health Plan**  
Policy ID: POL-2024-001

Download Reminder

ACTIVE

Coverage Amount  
**\$250,000**

Policy Holder  
**Alex Johnson**

Start Date  
**2024-01-01**

End Date  
**2026-12-31**

7  
Total Claims

2  
Pending

5  
Approved

**\$10,600**  
Total Claimed

#### Recent Claims

| Claim ID          | Description                             | Amount  | Status   | Submitted |
|-------------------|-----------------------------------------|---------|----------|-----------|
| CLM-1784195036605 | Annual physical exam                    | \$1,500 | Approved | 7/16/2026 |
| CLM-1784195024640 | Annual physical exam                    | \$1,500 | Approved | 7/16/2026 |
| CLM-1784194972793 | Annual physical exam                    | \$1,500 | Approved | 7/16/2026 |
| CLM-1721041800000 | X-ray imaging for dental work           | \$750   | Pending  | 7/15/2024 |
| CLM-1720170000000 | Physical therapy sessions (10 sessions) | \$3,000 | Pending  | 7/5/2024  |

Figure 1: POL-2024-001 (Alex Johnson) displays the prominent Orange Warning Banner with instructions and download buttons.

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DashboardClaims

### Policy Dashboard

6 policies on your account

Alex JohnsonPOL-2024-001

**Maria GarciaPOL-2024-002**

David ChenPOL-2023-009

Jane Doe

Download Reminder

ACTIVE

Coverage Amount  
**\$150,000**

Policy Holder  
**Maria Garcia**

Start Date  
**2024-02-15**

End Date  
**2027-02-14**

3  
Total Claims

1  
Pending

1  
Approved

**\$3,900**  
Total Claimed

#### Recent Claims

| Claim ID          | Description                 | Amount  | Status   | Submitted |
|-------------------|-----------------------------|---------|----------|-----------|
| CLM-1720598400000 | Dental cleaning and checkup | \$1,200 | Rejected | 7/10/2024 |
| CLM-1717955400000 | Prescription medications    | \$500   | Approved | 6/10/2024 |
| CLM-1717264800000 | Emergency room visit        | \$2,200 | Pending  | 6/1/2024  |

Figure 2: POL-2024-002 (Maria Garcia) does not trigger the warning banner, but shows the 'Download Reminder' action in the card.

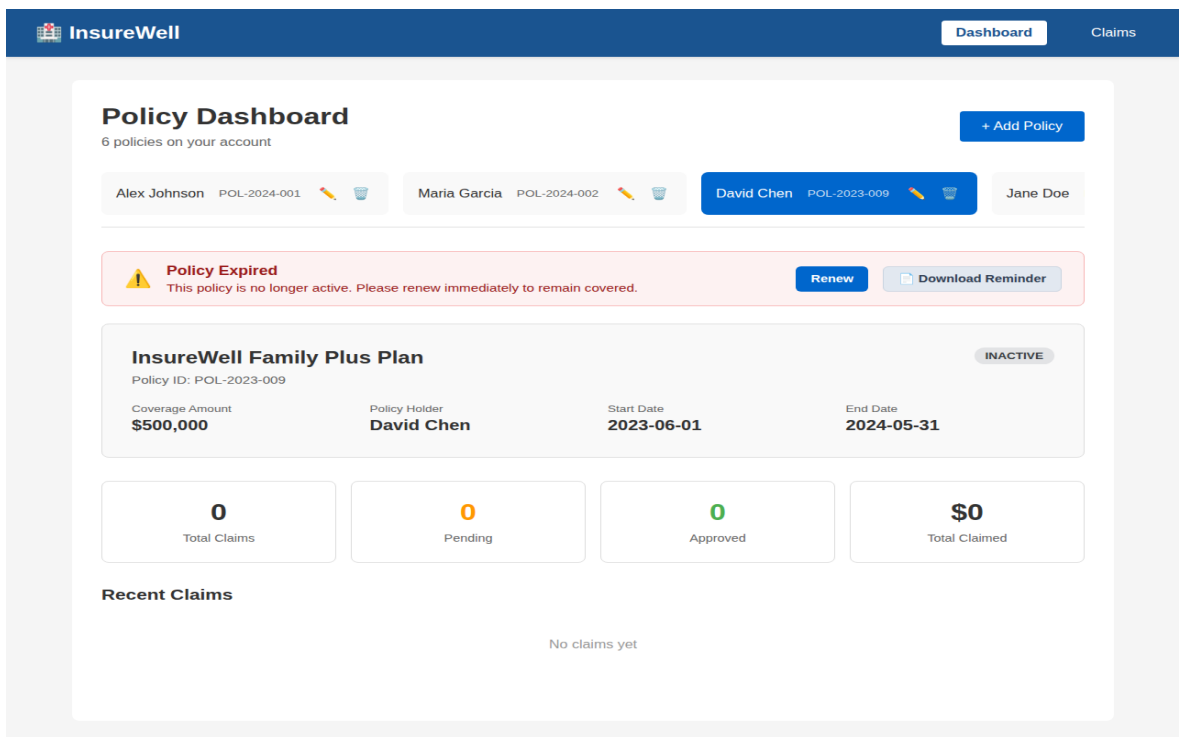


Figure 3: POL-2023-009 (David Chen) triggers the urgent Red Expired Banner, with download disabled on the inactive policy card.

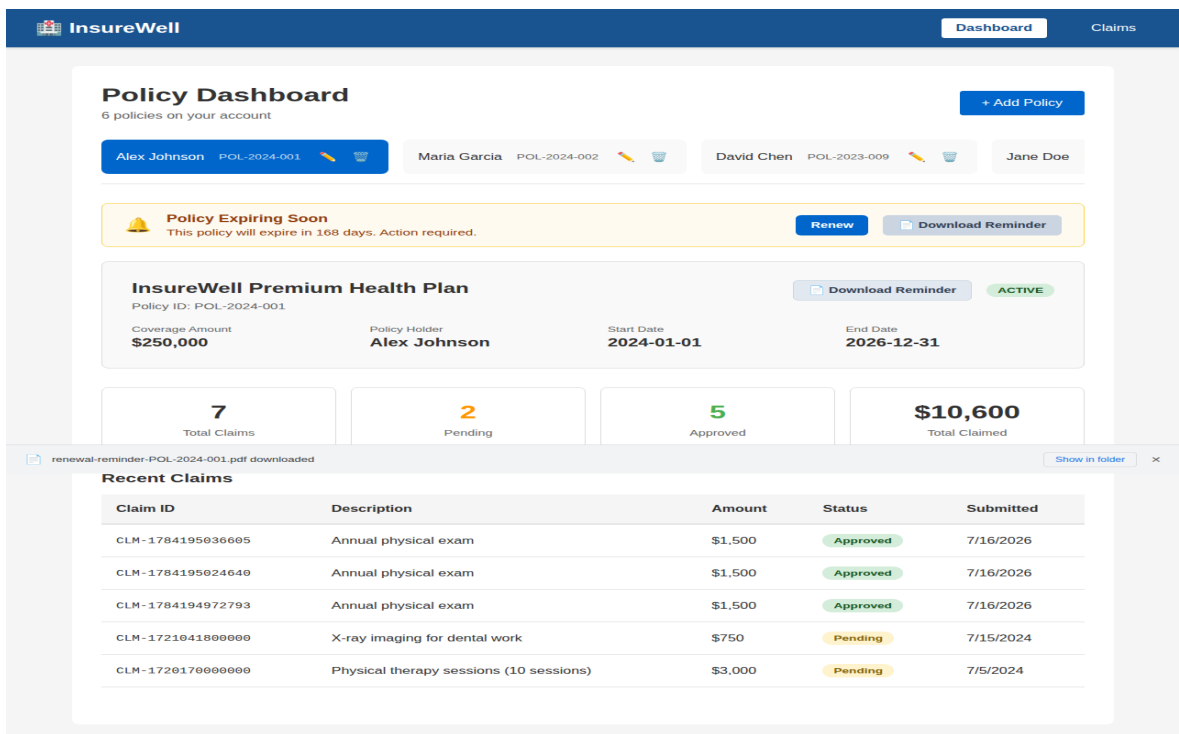


Figure 4: Simulated browser download shelf showing successful dynamic generation and download of the reminder PDF.

Generated Dynamic PDF Inspection

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Policy Renewal Reminder

|                   |                                |
|-------------------|--------------------------------|
|                   |                                |
| Policy ID         | POL-2024-001                   |
| Holder Name       | Alex Johnson                   |
| Plan Name         | InsureWell Premium Health Plan |
| Coverage Amount   | \$250,000.00                   |
| Start Date        | 2024-01-01                     |
| End Date          | 2026-12-31                     |
| Days Until Expiry | 168                            |

Your policy expires on 2026-12-31. Please contact InsureWell to renew before this date.

Generated at (UTC): 2026-07-16T09:50:36.898422156Z

Figure 5: Inspecting the generated PDF output rendered via PDFBox, including standard layout, metadata, tables, and brand coloring.